6CCSACCM
Cooperation, Coordination and Multi-Agent Systems

Week 4

Cooperative games (part I)

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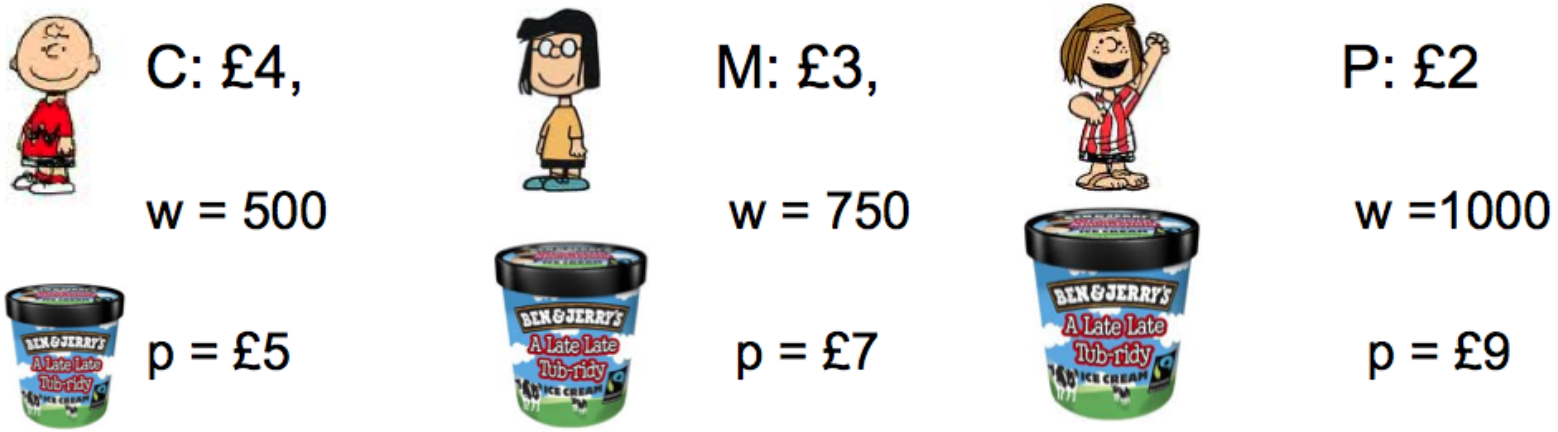
TU games formalised

A **transferable utility game** is a pair (N, v), where:

- $N = \{1, \ldots, n\}$ is a set of players.
- $v : 2^N \to \mathbb{R}$ is a characteristic function.
  - For each subset of players C, $v(C)$ is the amount that the members of C can earn by working together.
- Usually it is assumed that v is
  - *normalised*: $v(\emptyset) = 0$;
  - *non-negative*: $v(C) \geq 0$ for any $C \subseteq N$;
  - *monotone*: $v(C) \leq v(D)$ for any C, D such that $C \subseteq D$.
- A *coalition* is any subset of N.
  - N itself is called the *grand coalition*.

Example: ice-cream game



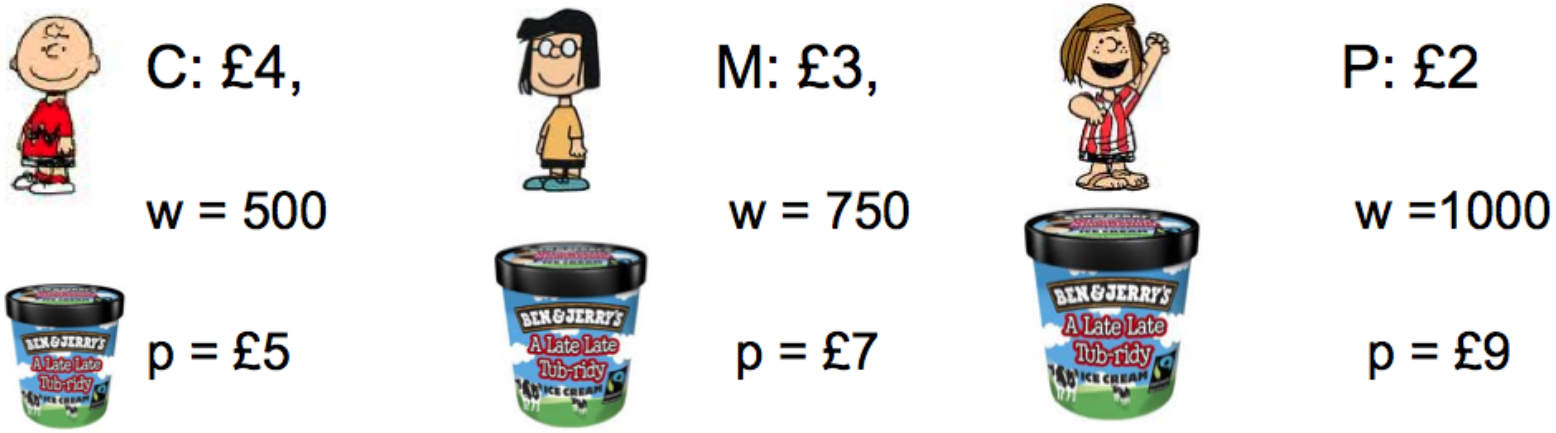
- $v(\emptyset) = v(\{C\}) = v(\{M\}) = v(\{P\}) = 0$
- $v(\{C, P\}) = v(\{M, P\}) = 500$, $v(\{C, M\}) = 750$
- $v(\{C, M, P\}) = 1000$

TU games: outcome

An **outcome** of a TU game (N, v) is a pair (CS, x), where:

- $CS = (C_1, \ldots, C_k)$ is a *coalition structure* – a partition of N into coalitions:
  - $\cup_{i=1}^{k} C_i = N$ and $C_i \cap C_j = \emptyset$ for $j \neq j$;
- $x = (x_1, \ldots, x_n)$ is a *payoff vector*, which distributes the value of each coalition in CS:
  - $x_i \geq 0$ for all $i \in N$;
  - $\sum_{i \in C} x_i = v(C)$ for each C in CS.
- Notation: we will denote $\sum_{i \in C} x_i$ by $x(C)$.

Example: ice-cream game



- $CS = (\{C\}, \{M\}, \{P\})$; $x_C = 0, x_M = 0, x_P = 0$
- $CS = (\{C\}, \{M, P\})$; $x_C = 0, x_M = 150, x_P = 350$
- $CS = (\{C, M, P\})$; $x_C = 334, x_M = 333, x_P = 333$
- $CS = (\{C, M, P\})$; $x_C = 0, x_M = 800, x_P = 200$

Example: weighted voting games

- n players with weights $w_1, \ldots, w_n$, quota q;
- $w(C) = \sum_{i \in C} w_i$;
- $v(C) = 1$ (C wins) if $w(C) \geq q$; $v(C) = 0$ (C loses) otherwise.

Weighted voting games

United Kingdom, 2010:

- 650 seats, $q = 326$
  - Conseratives (C): 306
  - Labour (L): 258
  - Liberal Democrats (LD): 57
  - Scottish National Party (SNP): 6
  - Democratic Unionist Party (DUP): 8
  - 6 other parties (O), with a total of 15 seats
- $N = \{C, L, LD, SNP, DUP, O\}$
- $v(\{C, L\}) = v(\{C, LD\}) = v(\{C, SNP, DUP, O\}) = 1$
- $v(\{L, LD\}) = 0$, $v(\{L, LD, SNP, DUP\}) = 1$

Example: network flow games

- network $G = (V, E)$, with source s and sink t, and edge capacities $c_1, \ldots, c_{|E|}$;
- players are edges;
- $v(C)$ is the maximum $s - t$ flow that C can carry.

Example: graph induced games

- players are vertices of a weighted graph
- value of a coalition = total weight of internal edges
- model for social networks
- weights need not be positive!
- *outcome example*: grand coalition forms, each player receives half of the weight of adjacent edges

Special classes: simple games

A game $G = (N, v)$ is **simple** if:

- value of each C is 0 or 1: $v(C) \in \{0, 1\}$ for any $C \subseteq N$
- v is *monotone*: if $S \subseteq T$ then $v(S) \leq v(T)$.
  - A coalition C in a simple game is said to be *winning* if $v(C) = 1$ and *losing* if $v(C) = 0$.
  - *Example:* weighted voting games with non-negative weights (but not general weighted voting games).

<u>Convention:</u> in simple games, unless specified otherwise, we assume that players form the *grand coalition*.

Simple games: player types

- In a simple game, a player $i \in N$ is a **null** player if: $v(C) = v(C \cup \{i\})$ for any $C \subseteq N \setminus \{i\}$.
- In a simple game, a player $i \in N$ is a **veto** player if: $v(C) = 0$ for any $C \subseteq N \setminus \{i\}$.
  - Equivalently, by monotonicity, $v(N \setminus \{i\}) = 0$.

Weighted voting games

United Kingdom, 2010:

- 650 seats, $q = 326$
  - Conseratives (C): 306
  - Labour (L): 258
  - Liberal Democrats (LD): 57
  - Scottish National Party (SNP): 6
  - Democratic Unionist Party (DUP): 8
  - 6 other parties (O), with a total of 15 seats
- no player is a null player
- no player is a veto player

Weighted voting games

United Kingdom, 2019:

- 650 seats, $q = 326$
  - Conseratives (C): 365
  - Labour (L): 202
  - Liberal Democrats (LD): 11
  - Scottish National Party (SNP): 47
  - Democratic Unionist Party (DUP): 8
  - 6 other parties + speaker(O), with a total of 17 seats
- $N = \{C, L, LD, SNP, DUP, O\}$
- C is a veto player, all others are null players

Special classes: superadditive games

A game $G = (N, v)$ is **superadditive** if $v(S \cup T) \geq v(S) + v(T)$ for any disjoint $S, T \subseteq N$.

- Two coalitions can always merge without losing money; hence, we can assume that players form the grand coalition.
- *Example:* Network flow games.

Superadditive games

- Convention: in superadditive games, we identify outcomes with payoff vectors for the grand coalition.
  - An outcome is a vector $x = (x_1, \ldots, x_n)$ with $\sum_{i \in N} x_i = v(N)$.
- Caution: some literature defines outcomes in this way even if the game is not superadditive.

Special classes: convex games

A function $f : 2^N \to \mathbb{R}$ is called **supermodular** if $f(\emptyset) = 0$ and $f(A \cup B) + f(A \cap B) \geq f(A) + f(B)$ for any $A, B \subseteq N$ (not necessarily disjoint).

- Any supermodular function is superadditive, but the converse is not true.

A TU game $G = (N, v)$ is **convex** if its characteristic function is supermodular.

Special classes: convex games

Proposition

If f is supermodular, $T \subset S$ and $i \notin S$ then
$f(T \cup \{i\}) - f(T) \leq f(S \cup \{i\}) - f(S)$.

- A player is more useful when he joins a bigger coalition.

Proof.

Let $A = S$ and $B = T \cup \{i\}$. Then, $A \cup B = S \cup \{i\}$ and $A \cap B = T$. $\square$

Ice-cream games: properties

Are ice-cream games superadditive?

- Yes.

Are ice-cream games convex?

- No.

Graph-induced games: properties

Are graph-induced games superadditive?

- If and only if the weight of each edge is non-negative.

Are graph-induced games convex?

- If and only if the weight of each edge is non-negative.

Weighted voting games: properties

Are weighted voting games superadditive?

- No (in general): $N = \{1, 2\}$; $w_1 = w_2 = 1$; $q = 1$.
- Yes, if $q > \frac{w(N)}{2}$

Are weighted voting games convex?

- Not necessarily, even if $q > \frac{w(N)}{2}$.
- *Example:* $N = \{1, 2, 3\}$; $w_1 = w_2 = w_3 = 1$; $q = 2$;

$$v(\{1, 2\}) + v(\{2, 3\}) = 2 > 1 = v(\{1, 2, 3\}) + v(\{2\}).$$

What is a good outcome?

C: £4,

w = 500

p = £5

M: £3,

w = 750

p = £7

P: £2

w =1000

p = £9

- $v(\emptyset) = v(\{C\}) = v(\{M\}) = v(\{P\}) = 0$
- $v(\{C, P\}) = v(\{M, P\}) = 500$, $v(\{C, M\}) = 750$
- $v(\{C, M, P\}) = 1000$

What is a good outcome?

C: £4,

w = 500

p = £5

M: £3,

w = 750

p = £7

P: £2

w =1000

p = £9

- This is a superadditive game.
  - Outcomes are payoff vectors.
- How should the players share the ice-cream?
  - If they share as $(350, 350, 300)$, C and M can get more ice-cream by buying a 750g tub on their own, and splitting it equally.
  - $\Rightarrow$ The outcome (350, 350, 300) is not stable!

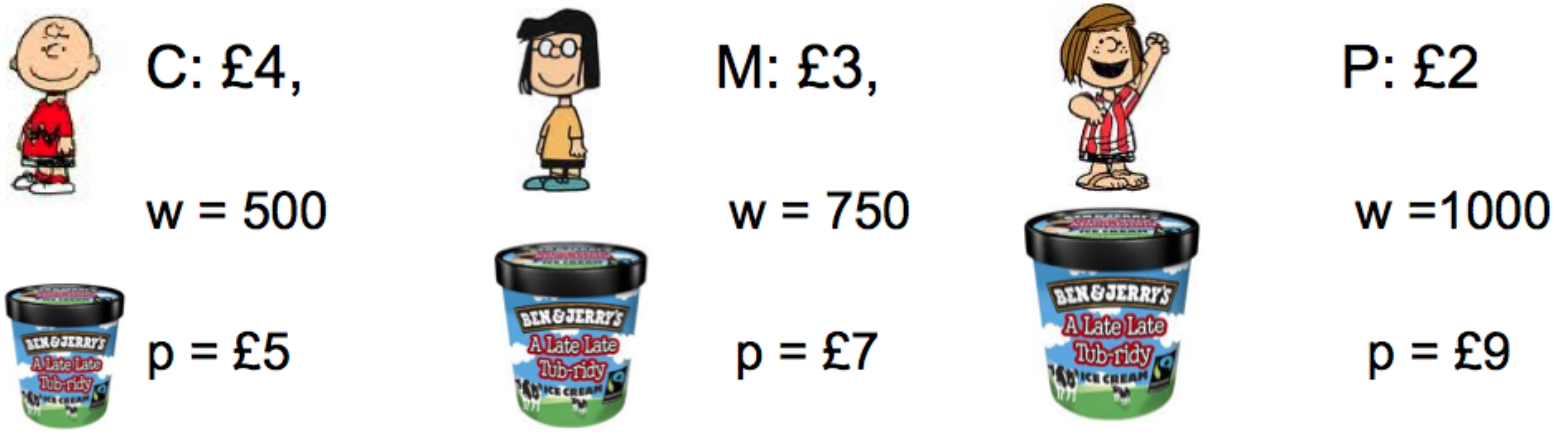
Core and core-related concepts

Definition

The core $\mathcal{C}(G)$ of a characteristic function game $G = (N, v)$ is the set of all outcomes $(CS, \mathbf{x})$ such that $x(C) \geq v(C)$ for every $C \subseteq N$.

- Each coalition earns at least as much as it can make on its own.
- Note that G is not assumed to be superadditive.

Example: ice-cream game



- $v(\emptyset) = v(\{C\}) = v(\{M\}) = v(\{P\}) = 0$
- $v(\{C, P\}) = v(\{M, P\}) = 500$, $v(\{C, M\}) = 750$
- $v(\{C, M, P\}) = 1000$

Example: ice-cream game

For any outcome $(CS, (x_C, x_M, x_P))$ in the core, it has to be $CS = \{N\}$ (since the game is super-additive).

- Otherwise, we would have

$$x(N) = \sum_{C \in \mathcal{CS}} v(C) < v(N)$$

violating the core constraint for N.

Example: ice-cream game

Furthermore, the values x_C, x_M, x_P have to satisfy the following constraints:

$$
\begin{aligned}
x_C \geq 0, \quad x_M \geq 0, \quad x_P &\geq 0 \\
x_C + x_M + x_P &= 1000 \\
x_C + x_M &\geq 750 \\
x_C + x_P &\geq 500 \\
x_M + x_P &\geq 500
\end{aligned}
$$

The system has several solutions. For example:

$$
\begin{aligned}
&(400, 400, 200) \\
&(500, 400, 100) \\
&(500, 500, 0)
\end{aligned}
$$

Core stability implies efficiency

- The outcomes in the core are stable in the sense that no coalition has incentives to deviate as it cannot do better on its own.
  - Therefore, such outcomes are more likely to arise when a coalitional game is played.
- Stability implies efficiency:
  - any outcome in the core maximises the *social welfare* – the total payoff to all the players.

Example: empty core

Recall the majority game with 3 players: WVG$[2; 1, 1, 1]$

- $v(C) = 1$ for $|C| \geq 2$;
- $v(C) = 0$ for $|C| \leq 1$.

Consider an outcome $(CS, \mathbf{x})$:

- if $CS = \{\{1\}, \{2\}, \{3\}\}$, the grand coalition can deviate;
- if $CS = \{\{1, 2\}, \{3\}\}$, either 1 or 2 gets payoff less than 1, so can deviate with player 3;
- same argument for $CS = \{\{1, 3\}, \{3\}\}$ and $CS = \{\{2, 3\}, \{1\}\}$;
- suppose $CS = \{\{1, 2, 3\}\}$: then, $x_i > 0$ for some i, so $x(N \setminus \{i\}) < 1$, yet $v(N \setminus \{i\}) = 1$.

The core of convex games

Proposition

Any convex game has a non-empty core.

Proof.

Set $x_1 = v(\{1\})$, $x_2 = v(\{1,2\}) - v(\{1\})$, ..., $x_n = v(\{N\}) - v(N \setminus \{n\})$, i.e., pay each player its marginal contribution to the coalition formed by its predecessors. We have $x_1 + x_2 + \cdots + x_n = v(N)$, so $\mathbf{x}$ is a payoff vector. It remains to show that it is in the core.

Pick any coalition $C = \{i, j, \ldots, s\}$ where $i < j < \ldots < s$. We shall prove that $v(C) \leq x_i + x_j + \ldots + x_s$, i.e., C cannot deviate.

$$v(C) = v(\{i\}) + v(\{i,j\}) - v(\{i\}) + \ldots + v(\{C\}) - v(C \setminus \{s\})$$

- $v(\{i\}) = v(\{i\}) - v(\emptyset) \leq v(\{1, \ldots, i-1, i\}) - v(\{1, \ldots, i-1\}) = x_i$
- $v(\{i,j\}) - v(\{i\}) \leq v(\{1, \ldots, j-1, j\}) - v(\{1, \ldots, j-1\}) = x_j$
- $v(C) - v(C \setminus \{s\}) \leq v(\{1, \ldots, s-1, s\}) - v(\{1, \ldots, s-1\}) = x_s$

$\square$

The core of convex games

This proof suggests a simple algorithm
for constructing an outcome in the core:

- order the players as $1, \ldots, n$;
- compute $v(\{1\}), v(\{1,2\}), \ldots, v(N)$;
- set $x_i = v(\{1, \ldots, i-1, i\}) - v(\{1, \ldots, i-1\})$.

I.e., for convex games an outcome in the core can be constructed in time polynomial in n given oracle access to the characteristic function,

- even though the representation of the game involves 2^n numbers.

The core of simple games

<u>Convention:</u> in simple games we always assume that the grand coalition forms

- outcome = payoff vector for the grand coalition

Theorem

A simple game has a non-empty core iff it has a veto player.

Proof of theorem

$\Leftarrow$

- The case $v(N) = 0$ is easy, so suppose $v(N) = 1$.
- Suppose i is a veto player.
- Consider a payoff vector $\mathbf{x}$ with $x_i = 1$, $x_k = 0$ for $k \neq i$.
- No coalition C can deviate from $\mathbf{x}$:
  - if $i \in C$, we have $x(C) = 1 \geq v(C)$;
  - if $i \notin C$, we have $v(C) = 0$.

Proof of theorem

$\Rightarrow$

- The case $v(N) = 0$ is easy, so suppose $v(N) = 1$.
- Suppose there is no veto player.
- Consider an arbitrary payoff vector $\mathbf{x}$:
  - we have $x(N) = v(N) = 1$, thus $x_i > 0$ for some $i \in N$;
  - but then $N \setminus \{i\}$ can deviate:
    since i is not a veto, $v(N \setminus \{i\}) = 1$,
    yet $x(N \setminus \{i\}) = 1 - x_i < 1$. $\square$

The core of simple games

Corollary

In a simple game G, a payoff vector $\mathbf{x}$ is in the core iff $x_i = 0$ for any non-veto player i.

- proved similarly

Checking if a player i is a veto player is easy:

- a single query to compute $v(N \setminus \{i\})$

Thus, in simple games, checking *non-emptiness of the core* or checking *if a given outcome is in the core* is easy given oracle access to the characteristic function.

Simple games with coalition structures

What if we remove the assumption that in simple games the grand coalition forms?

- Then there may exist core stable outcomes even if there are no veto players:
  - *Example:* $G = [2; 1, 1, 1, 1]$
  - $(\{\{1, 2\}, \{3, 4\}\}, (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}))$ is in the core
  - if $|C| = 1$ then $v(C) = 0$, and if $|C| > 1$ then $x(C) \geq 1 = v(C)$.

Can we then efficiently decide if the core is non-empty?

Weighted voting games with coalition structures

Theorem

If we allow coalition structures in weighted voting games, deciding if the core is non-empty is NP-hard.

Proof: reduction from PARTITION.

<u>PARTITION:</u>

- Instance: list of positive integers $A = (a_1, \ldots, a_n)$ given in binary that sum up to $2K$, $a_i < K$ for each i
- Question: can we split A into A' and A'' so that both A' and A'' sum up to K?

Proof of theorem

Reduction: $(a_1, \ldots, a_n; 2K) \to$ WVG $G = [K; a_1, \ldots, a_n]$.

- if there is a partition of A as A', A'' that each sum to K:
  - partition the players in G accordingly;
  - pay each player in proportion to their weight: $x_i = \frac{a_i}{K}$;
  - stability: if C can deviate, then $a(C) \geq K$, but then $x(C) \geq 1 = v(C)$.
- if there is no partition:
  - there is at most one winning coalition;
  - fix an outcome $(CS, \mathbf{x})$ and let i be a player with $x_i > 0$;
  - then the set of all players other than i can deviate to form a winning coalition.

The core of superadditive games

For superadditive games we can simplify the notation and define the core as the set of all vectors $\mathbf{x}$ that satisfy:

- $x_i \geq 0$ for all $i \in N$;
- $x(N) = v(N)$;
- $x(C) \geq v(C)$ for all $C \subseteq N$.

Example: the core of NOT superadditive games

If the game is not superadditive, it may be the case that the core of the game is non-empty, even though no outcome in which the grand coalition forms is stable.

- $N = \{1, 2, 3, 4\}$
- $v(C) = 0$ if $|C| < 2$
- $v(C) = 2$ if $|C| \geq 2$

It is easy to check that the outcome $(\{\{1, 2\}, \{3, 4\}\}, (1, 1, 1, 1))$ is in the core, while no outcome of the form $(\{N\}, \mathbf{x})$ is in the core, since there's a pair of agents that are paid less than 2 and therefore want to deviate.

Superadditive cover

Any non-superadditive game can be transformed into a superadditive game by computing, for each coalition, the maximum amount this coalition can earn by splitting into subcoalitions.

- Given a (non-superadditive) game $G = (N, v)$, we can define a new game $G^* = (N, v^*)$ by setting

$$v^*(C) = \max_{CS \in \mathcal{CS}_C} v(CS)$$

  for every coalition $C \subseteq N$, where $\mathcal{CS}_C$ denotes the space of all coalition structures over C.

Superadditive cover

Theorem

A characteristic function game $G = (N, v)$ has a non-empty core if and only if its superadditive cover $G^* = (N, v^*)$ has a non-empty core.

ε-core

- The notion of stability captured in the core requires that no coalition can gain *anything* by deviating.
- That is, an outcome is unstable if a coalition can benefit even by a tiny amount from deviating.
- In many situations, a deviation may be costly, and it would make sense for a coalition to only deviate if the gain from a deviation exceeded the costs of deviation.
- We can relax the notion of the core by requiring that no coalition can benefit *significantly* by deviating.

ε-core

Definition

An outcome $\mathbf{x}$ is said to be in the ε-core of a superadditive game G for some $\varepsilon \geq 0$ if for each $C \subseteq N$,

$$x(C) \geq v(C) - \varepsilon$$

Least core

Definition

Given a superadditive game G, let

$$\varepsilon^*(G) = \inf\{\varepsilon | \varepsilon - \text{core of } G \text{ is non} - \text{empty}\}$$

The **least core** of G is its $\varepsilon^*(G)$-core. The quantity $\epsilon^*(G)$ is called the *value of the least core* of G.

Least core

By modifying the LP for the core of a superadditive game, we can obtain an LP for computing the value of the least core as well as an outcome in the least core; this proves that the least core is always non-empty.

$$
\begin{aligned}
&\min \varepsilon \\
x_i &\geq 0 \quad \text{for all} \quad i \in N \\
x(N) &= v(N) \\
x(C) &\geq v(C) - \varepsilon \quad \text{for all} \quad C \subseteq N
\end{aligned}
$$

Cost of stability

Definition

Given a superadditive game $G = (N, v)$ and a $\Delta \geq 0$, let $G^\Delta = (N, v^\Delta)$ be a game over the set of players N given by $v^\Delta(N) = v(N) + \Delta$ and $v^\Delta(C) = v(C)$ for all $C \subsetneq N$.

The **cost of stability** of G is the smallest (nonnegative) value of Δ such that G^Δ has a non-empty core.

Cost of stability

Similarly, we can design a linear program to compute the cost of stability:

$$
\begin{aligned}
\min \Delta & \\
x_i &\geq 0 \quad \text{for all} \quad i \in N \\
x(N) &= v(N) + \Delta \\
x(C) &\geq v(C) \quad \text{for all} \quad C \subseteq N
\end{aligned}
$$

Shapley value

- One of the solution concepts that capture the notion of fairness is the **Shapley value**.
- The Shapley value is based on the intuition that the payment that each agent receives should be *proportional to its contribution*.

Shapley value

- Let Π_N denote the set of all permutations of N.
- Given a permutation $\pi \in \Pi_N$, let $S_\pi(i)$ be the set of all predecessors of i in π:

$$S_\pi(i) = \{j \in N | \pi(j) < \pi(i)\}$$

Shapley value

The **marginal contribution** of an agent i with respect to a permutation π in a game $G = (N, v)$ is given by

$$\Delta_\pi^G(i) = v\left(S_\pi(i) \cup \{i\}\right) - v\left(S_\pi(i)\right)$$

Definition

Given a characteristic function game $G = (N, v)$ with $|N| = n$, the **Shapley value** of a player $i \in N$ is given by

$$\phi_i(G) = \frac{1}{n!} \sum_{\pi \in \Pi_N} \Delta_\pi^G(i)$$

Example: ice-cream game

- Let's calculate the Shapley value for Pattie.
- There are six permutations of the players:

$$\pi_1 = CMP \quad \pi_2 = MCP \quad \pi_3 = CPM$$
$$\pi_4 = MPC \quad \pi_5 = PCM \quad \pi_6 = PMC$$

- We have:

$$\begin{aligned}
\Delta^G_{\pi_1}(i) = \Delta^G_{\pi_2}(i) &= v(N) - v(\{C, M\}) = 250 \\
\Delta^G_{\pi_3}(i) &= v(\{C, P\}) - v(\{C\}) = 500 \\
\Delta^G_{\pi_4}(i) &= v(\{M, P\}) - v(\{M\}) = 500 \\
\Delta^G_{\pi_5}(i) = \Delta^G_{\pi_6}(i) &= v(\{P\}) - v(\emptyset) = 0
\end{aligned}$$

Hence, $\phi_P(G) = \frac{250+250+500+500}{6} = 250.$

Shapley value: properties

The Shapley value is **efficient**, i.e., it distributes the value of the grand coalition among all agents.

- For any characteristic function game $G = (N, v)$,

$$\sum_{i \in N} \phi_i(G) = v(N)$$

Shapley value: properties

Another useful property of the Shapley value is that it does not allocate any payoffs to players who do not contribute to any coalition.

- Given a characteristic function game $G = (N, v)$, a player i is said to be a **null player (or, dummy)** if $v(C \cup \{i\}) = v(C)$ for any $C \subseteq N$.
- If $i \in N$ is a dummy in G, then $\phi_i(G) = 0$.

Shapley value: properties

Another important observation is that if two players contribute equally to each coalition,then their Shapley values are equal.

- Given a characteristic function game $G = (N, v)$, we say that players i and j are **symmetric** in G if

$$v(C \cup \{i\}) = v(C \cup \{j\})$$

  for any $C \subseteq N \setminus \{i, j\}$.
- If players i and j are symmetric in G, then $\phi_i(G) = \phi_j(G)$.

Shapley value: properties

Consider a group of players N that is involved in two coalitional games G^1 and G^2:

- $G^1 = (N, v^1)$;
- $G^2 = (N, v^2)$.

The **sum** of G^1 and G^2 is a coalitional game $G^+ = G^1 + G^2$ given by $G^+ = (N, v^+)$, where for any coalition $C \subseteq N$,

$$v^+(C) = v^1(C) + v^2(C)$$

For any player i, we have

$$\phi_i(G^+) = \phi_i(G^1) + \phi_i(G^2)$$

Shapley value: properties

To summarise, we have shown that the Shapley value possesses four desirable properties:

1. **Efficiency**: $\sum_{i \in N} \phi_i(G) = v(N)$;
2. **Dummy player**: if $i \in N$ is a dummy, then $\phi_i(G) = 0$;
3. **Symmetry**: if i and j are symmetric, then $\phi_i(G) = \phi_j(G)$;
4. **Additivity**: $\phi_i(G^1 + G^2) = \phi_i(G^1) + \phi_i(G^2)$.

- Shapley value is the *only* payoff division scheme that has these four properties.
- These properties can also be used to simplify the computation of the Shapley value.

THANK YOU!